Exelon Generation.

SVP-20-080

10 CFR 50.73

November 10, 2020

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Unit 1
Renewed Facility Operating License No. DPR-28 and DPR-29
NRC Docket No. 50-254 and 50-265

Subject: Licensee Event Report 254/2020-002-00 "Main Steam Line Drain Valve Declared Inoperable Due to Improper Torque Switch Setting"

Enclosed is Licensee Event Report 254/2020-002-00 "Main Steam Line Drain Valve Declared Inoperable Due to Improper Torque Switch Setting," for Quad Cities Nuclear Power Station, Unit 1 and 2.

This report is submitted in accordance with 10 CFR 50.73(a)(2)(i)(B), for any operation or condition which was prohibited by the plant's Technical Specifications.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Sherrie Grant at (309) 227-2800.

Respectfully,

for Ken Ohr
Brian Wake

Kenneth S. Ohr
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station

LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Quad Cities Nuclear Power Station	05000-0254	1 OF 4

4. Title
Main Steam Line Drain Valve Declared Inoperable Due to Improper Torque Switch Setting

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
09	11	2020	2020	- 002 -	00	11	10	2020	Quad Cities	05000-0265
									Facility Name	Docket Number 05000

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Rachel Luebbe – Principal Regulatory Engineer	309-227-2813

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
A	SB	ISV	C665	Y	A	SB	ISV	C665	Y

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On July 1, 2020 and July 23, 2020, engineering identified the torque switch setting on the Unit 2 and Unit 1 Main Steam Line (MSL) outboard drain (1(2)-0220-2) motor operated valves (MOV) were found to be nonconservative with respect to the valve's design function under the expected worst-case operating conditions. Both valves were initially evaluated and declared operable because the valve was closed (safety function position) and was not required to be opened during the normal two-year operating cycle. On September 11, 2020 at 1100 CDT, Operations determined that although the valves were performing their isolation function, operability was no longer supported and the MSL outboard drain valves were declared inoperable.

This issue is being reported under 10CFR50.73(a)(2)(i)(B) for any operation or condition which was prohibited by the plant's Technical Specifications (TS) because TS 3.6.1.3 Action Statement A.1 was not completed within the required time in July 2020 during the initial discovery of the torque switch setting issue.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Quad Cities	05000- 0254	2020	002	00

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric – Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

Main Steam Drain Valve Declared Inoperable Due to Improper Torque Switch Setting

A. CONDITION PRIOR TO EVENT

Unit: 1&2	Event Date: September 11, 2020	Event Time: 1100
Reactor Mode: 1	Mode Name: Power Operations	Power Level: 100%

There were no other structures, systems or components (SSC) inoperable during this event time period that could have contributed to this event.

B. DESCRIPTION OF EVENT

On July 1, 2020, following review of an industry operating experience report, engineering identified the torque switch setting on the Unit 2 Main Steam [SB] Line outboard drain valve [ISV] (2-0220-2) motor operated valve (MOV) was nonconservative with respect to its design function under the expected worst-case operating conditions. The cause of as-left torque switch setting being lower than the required value was determined to be a latent error in the calculation used for these valves. The valve was initially called operable with a request for a formal Operability Evaluation (Op Eval), because the valve was closed (safety function position) and was not required to be opened during the normal two-year operating cycle. Revision 00 of the Op Eval, completed on July 7, 2020, provided compensatory actions for the degraded automatic isolation function by revising procedures to have a dedicated operator ready to enter the Main Steam Isolation Valve (MSIV) room to manually ensure full closure of the valve should the valve need to be opened.

On July 23,2020, following an extent of condition review, engineering identified that the Unit 1 Main Steam Line outboard drain valve (1-0220-2) MOV torque switch setting was also incorrect. The Op Eval that was written to cover the previously identified Unit 2 valve was revised on August 5, 2020 to include the Unit 1 valve and requested similar compensatory actions. The basis for the Unit 2 valve operability was also applied to the Unit 1 valve.

On September 11, 2020 at 1100 CDT, Operations determined that the Op Eval no longer supported operability of the Unit 1 and Unit 2 Main Steam Line outboard drain valves because of an inability to meet Primary Containment Isolation Valve (PCIV) TS surveillance requirements (SR). The affected valves were declared inoperable and Technical Specification 3.6.1.3 Condition A was entered at that time. On

NARRATIVE

September 11, 2020 at 1339 CDT, the affected penetration flow path was isolated in accordance with TS 3.6.1.3 Action Statement A.1.

It was determined the valves had been inoperable since the initial discovery of the torque switch trip setting issue in July 2020. Thus, this issue is being reported under 10 CFR 50.73(a)(2)(i)(B) for any operation or condition which was prohibited by the plant's Technical Specifications.

C. CAUSE OF EVENT

The cause of the Main Steam Line Drain MOVs inoperability was an incorrect torque switch setting due to historical non-conservatisms in the governing calculation. The condition prohibited by TS was caused by an operability determination that was not in alignment with the TS SRs regarding PCIV operability.

D. SAFETY ANALYSIS

System Design

The Main Steam Line Drain Valves are considered automatic PCIVs. The function of the PCIVs, in combination with other accident mitigation systems, is to limit fission product release during and following postulated Design Basis Accidents (DBAs). Primary Containment isolation within the time limits specified for those isolation valves designed to close automatically ensures that the release of radioactive material to the environment will be consistent with the assumptions using the safety analyses for a DBA. The MSL drain valves are normally closed and not required to be opened during the normal operating cycle.

Safety Impact

Both Unit 1 and Unit 2 Main Steam Line Drain Valves have been fully closed, which is their safety function position, since Unit 1 and Unit 2 have been at power operation following their respective refueling outages. Since the valves are not opened during normal unit operation, the safety consequences of the degraded condition which only affects the valves ability to re-close after opening is minimal.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
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E. CORRECTIVE ACTIONS

The Main Steam Line drain valves will be repaired to correct the torque switch setting issue in the next refuel outage on each unit.

Follow-up corrective actions include additional training for both Operations and Engineering on the Operability Evaluation process and an extent of condition review on all open Operability Evaluations to ensure conclusions support continued operability.

F. PREVIOUS OCCURRENCES

The station events database, LERs and IRIS were reviewed for similar events at Quad Cities Nuclear Power Station, with one event being identified. This event was a Safe Shutdown Makeup Pump (SSMP) discharge pressure calculation error. This event is similar in that it resulted in a condition outside of Technical Specification that was related to an engineering error in a calculation.

LER (254/2008-001-00) Past Operation of Safe Shutdown Makeup Pump Outside Technical Specifications Surveillance Requirements, 09/04/2008. An error in a calculation found that SSMP discharge pressure necessary to meet the TS surveillance requirement (SR) had increased. This calculation error resulted in historical TS SR testing failures of the corrected required SSMP discharge pressure. The cause was determined to be lack of procedural guidance in the performance and review of calculations.

G. COMPONENT FAILURE DATA

Failed Equipment: Unit 1 and Unit 2 Main Steam Line outboard drain valve MOV
Component Manufacturer: Crane
Component Model Number: 783-U
Component Part Number: N/A

This event has been reported to IRIS.